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CS-210: Programming Languages

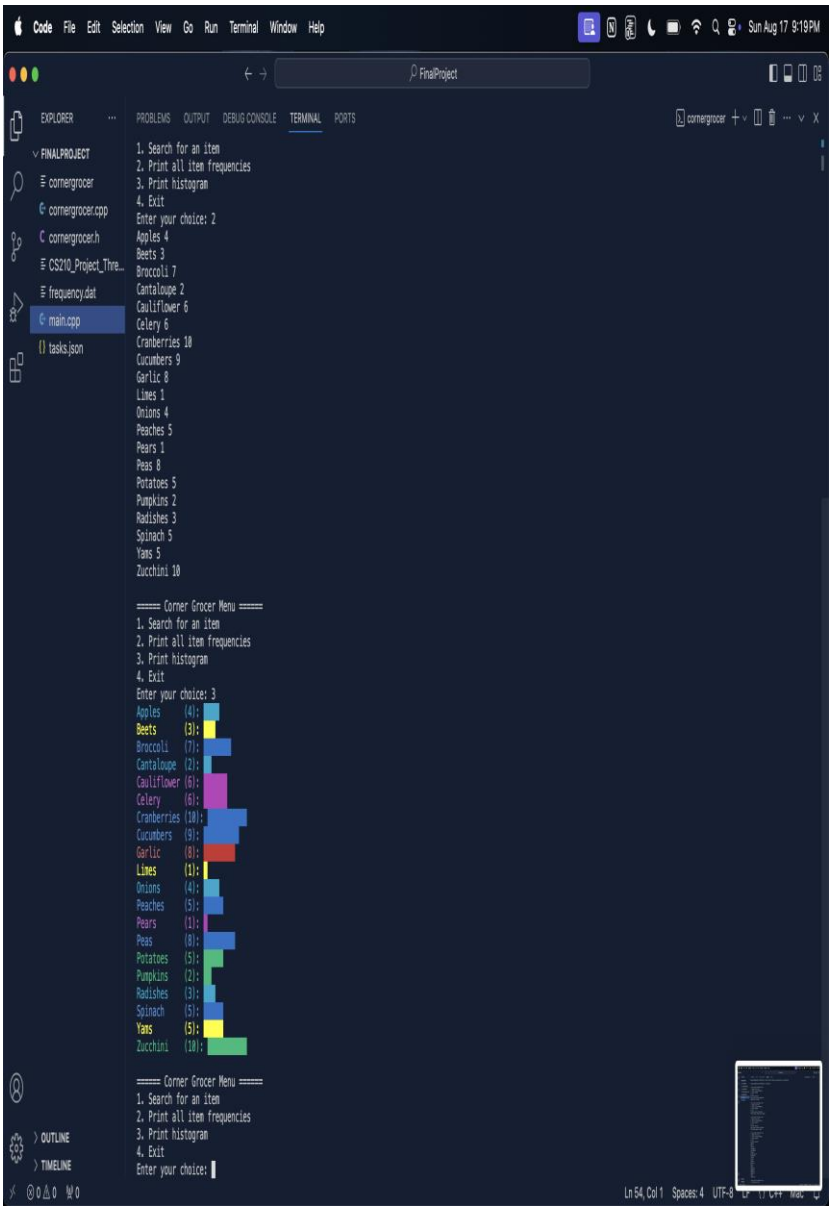
Project Three: Documentation

The Corner Grocer Program is a C++ console application designed to track grocery store purchases. It reads an input file containing a list of items purchased during the day and counts how many times each item appears. The program stores these counts in a map and automatically creates a backup file called frequency.dat to save the results. Users interact with the program through a menu that allows them to search for a specific item, view a complete list of item frequencies, or view a histogram. The histogram displays each item with its count in parentheses and a bar made of " " characters. Each item is randomly assigned a color to make the output more visually distinctive, and the bars are aligned so they all start at the same column regardless of the length of the item name.

The program is implemented using a CornerGrocer class that encapsulates all functionality. The class has private data to store item frequencies, and the ANSI color codes for the histogram. Public methods include loading data from a file, saving the frequencies to a backup file, searching for a single item, printing all item counts, and printing the histogram. The program uses standard C++ libraries including <map>, <string>, <fstream>, <cstdlib>, and <ctime>. Inline comments were included throughout the code to explain the functionality of each section. Error handling is implemented for file input to ensure the program notifies the user if a file cannot be opened.

Screenshots have been included to show the menu, a sample search result, the full frequency list, and the histogram with colored bars. The program was designed using modern

C++ practices and object-oriented principles to make it easy to maintain and extend.


Code File Edit Selection View Go Run Terminal Window Help
Sun Aug 17 9:19PM
FinalProject
EXPLORER
FINALPROJECT
cornergrocer
cornergrocer.cpp
cornergrocer.h
CS210_Project_Thre...
frequency.dat
main.cpp
tasks.json
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
cornergrocer
1. Search for an item
2. Print all item frequencies
3. Print histogram
4. Exit
Enter your choice: 2
Apples 4
Beets 3
Broccoli 7
Cantaloupe 2
Cauliflower 6
Celery 6
Cranberries 10
Cucumbers 9
Garlic 8
Limes 1
Onions 4
Peaches 5
Pears 1
Peas 8
Potatoes 5
Pumpkins 2
Radishes 3
Spinach 5
Yams 5
Zucchini 10
==== Corner Grocer Menu ====
1. Search for an item
2. Print all item frequencies
3. Print histogram
4. Exit
Enter your choice: 3
Apples (4):
Beets (3):
Broccoli (7):
Cantaloupe (2):
Cauliflower (6):
Celery (6):
Cranberries (10):
Cucumbers (9):
Garlic (8):
Limes (1):
Onions (4):
Peaches (5):
Pears (1):
Peas (8):
Potatoes (5):
Pumpkins (2):
Radishes (3):
Spinach (5):
Yams (5):
Zucchini (10):
==== Corner Grocer Menu ====
1. Search for an item
2. Print all item frequencies
3. Print histogram
4. Exit
Enter your choice:
OUTLINE
TIMELINE
Ln 54, Col 1 Spaces: 4 UTF-8

References:

Macha. (2010, April 11). *How do I output coloured text to a Linux terminal?* Stack Overflow. https://stackoverflow.com/questions/2616906/how-do-i-output-coloured-text-to-a-linux-terminal